



Ignitron IPT 2.0 Firmware Changes

Complete engineering and operating record for the customized Ignitron firmware, including USB preset tooling, filesystem transfer, live display integration, larger OLED support, AMP-mode hardware control, and pedal-only Spark amp blocking.

Document basis

Prepared from the firmware tree at `T:\ignitron preset tools v2.0\Ignitron`. Firmware version string: **2.0.1**. Build verified on July 20, 2026 using PlatformIO for the node32s environment.

Scope

This report records the changes present in the current IPT 2.0 firmware and the pedal blacklist feature added during the current work. It is a change summary and operating reference, not a replacement for the upstream Ignitron build and wiring documentation.

Contents

1. Executive summary
2. USB serial protocol
3. Preset puller and Spark App capture
4. Filesystem write and preset selection
5. Live display, tuner stream, and bigger-screen support
6. AMP-mode rocker switch
7. Pedal-only unwanted-amp blacklist
8. Build, filesystem, and partition configuration
9. Changed-file inventory and verification

1. Executive summary

IPT 2.0 turns Ignitron into a two-way USB-managed preset device while retaining normal Spark control. The firmware can enumerate and export presets, receive validated files, select presets remotely, mirror the display, stream tuner values, and capture presets arriving from the Spark app.

Change area	What changed	User impact
USB command service	Line-buffered command parser runs during normal operation and connection waits.	IPT remains responsive even while the pedal is searching for an amp.
Preset access	Preset JSON and bank lists can be framed and returned over serial.	Backup and inspection no longer require direct flash extraction.
Safe file transfer	Base64 upload uses declared size and additive checksum validation.	IPT can update LittleFS files with explicit success/error responses.
Live telemetry	Display mirror and tuner streams have independent rate limits.	Desktop tools can reflect pedal state without polling the OLED hardware.
Display hardware	128x64 SSD1306/SSD1309/SH1106/SH1107 paths are selectable; current build selects SH1106.	Supports physically larger 1.3-inch or 2.42-inch modules while preserving the 128x64 layout.
Pedal controls	AMP-mode rocker switch and persistent blocked-amp MAC workflow.	Less button-holding at boot and no accidental reconnection to a nearby Spark.

Important distinction

The 'bigger screen' change is primarily a **controller/physical-size compatibility** change. The drawing canvas remains 128 x 64 pixels. The current configuration enables the SH1106 path, commonly used by 1.3-inch OLED modules; SSD1309 support is available for many 2.42-inch 128 x 64 modules.

2. USB serial protocol

The main sketch now mounts LittleFS before starting device services and calls `handleSerialCommands()` both in the main loop and while waiting for a BLE amp connection. Commands are newline terminated at 115200 baud. The parser normalizes command names to uppercase and limits its input buffer.

Command reference

Command	Purpose	Primary response
PING	Connection/liveness check	PONG
FSINFO	Report LittleFS capacity and usage	FSINFO total=... used=...
LISTFILES	Enumerate root-level files	LISTFILES_START / FILE ... / LISTFILES_DONE
LISTPRESETS	Return root-level JSON presets	LISTPRESETS_START / JSON STRING: ... / LISTPRESETS_DONE
LISTBANKS	Return /PresetList.txt	LISTBANKS_START / content / LISTBANKS_DONE
MIRROR ON OFF	Enable/disable display state stream	OK MIRROR ON or OFF
TUNERSTREAM ON OFF	Enable/disable tuner telemetry	OK TUNERSTREAM ON or OFF
SELECTPRESET b p	Select a bank and preset	OK or ERR SELECTPRESET b p
PUTBEGIN name size sum	Begin validated file transfer	Ready/error response
PUTB64 data	Append one Base64 transfer chunk	Chunk error only when invalid
PUTEND	Validate and commit transfer	OK or detailed validation error
PUTABORT	Cancel transfer and clean temporary state	Abort acknowledgement
DELETE name	Delete an allowed root file	OK DELETE or ERR DELETE_FAILED
RESTART	Restart ESP32	OK RESTART, then reboot

Protocol framing makes completion deterministic. IPT can stop reading at the matching `*_DONE` marker instead of relying on timeouts. JSON is emitted on one line with carriage returns, line feeds, and tabs removed for reliable desktop parsing.

3. Preset puller and Spark App capture

Preset pulling

The firmware scans the LittleFS root for case-insensitive `.json` files. For each match it prints the filename, then the complete preset as a single `JSON STRING: record`. Bank order is returned separately from `/PresetList.txt`. Together these records let IPT reconstruct the pedal's preset library and bank map.

Spark App capture

When Ignitron receives a preset from the Spark app in AMP mode, `SparkPresetControl::updateFromSparkResponseAmpPreset()` now writes the normalized preset JSON to the USB serial stream even when debug logging is disabled. IPT can detect the `received from app:` marker and save the following JSON.

Representative serial flow

```
LISTPRESETS
LISTPRESETS_START
Reading preset filename: /Rage.json
JSON STRING: { ... }
LISTPRESETS_DONE
```

Operating note
Spark App scraping depends on Ignitron operating as an amp/server to the app. The optional rocker switch described in section 6 makes entering AMP mode repeatable without holding a preset switch at every boot.

Files involved

File	Role
Ignitron.ino	Serial parser, preset/bank enumeration, transfer protocol, selection, restart.
src/SparkPresetControl.cpp	Always-on serial emission of presets received from the Spark app.
data/PresetList.txt	Ordered bank/preset mapping returned by LISTBANKS.
LittleFS root *.json	Stored preset documents returned by LISTPRESETS.

4. Filesystem write and preset selection

IPT 2.0 adds a transfer state machine rather than writing arbitrary serial data directly to flash. A transfer declares its destination filename, expected byte count, and additive checksum. Base64 chunks are decoded incrementally; completion succeeds only when the actual byte count and checksum match.

Safety behavior

- Filenames are normalized to the LittleFS root; empty names, nested paths, and . . traversal are rejected.
- Only one upload can be active at a time, with explicit abort and cleanup state.
- Malformed Base64, incorrect length, checksum mismatch, and filesystem errors return machine-readable errors.
- Delete operations use normalized root paths and report success or failure.
- Remote preset selection validates the requested bank and preset before changing the active selection.

Transfer sequence

Step	Desktop sends	Firmware action
1	PUTBEGIN Foo.json 1234 00ABCDEF	Open destination and initialize counters.
2	PUTB64 eyJWZXJzaW9u...	Decode and append bytes; update checksum.
3	PUTB64 ...	Repeat until all data is sent.
4	PUTEND	Close, validate length/checksum, return final result.

Flash-wear note

These commands write LittleFS. Normal preset-management usage is appropriate, but continuous high-frequency rewriting should be avoided.

5. Live display, tuner stream, and bigger-screen support

Live desktop integration

The display controller can emit a serial representation of the current four-line UI state. Display mirroring is rate-limited to approximately 250 ms; tuner telemetry is rate-limited to approximately 100 ms. The physical tuner also keeps tuner output active while its mode is selected.

Stream	Firmware source	Use
Display mirror	Current mode, preset name, bank/preset, secondary status	Reproduce pedal state in IPT without screen capture.
Tuner	Current note and cents offset	Responsive desktop tuner visualization.
Temporary messages	Direct OLED status page	Shows AMP BLOCKED, NO AMP CONNECTED, and AMP BLOCKLIST CLEARED.

Bigger OLED support

The display abstraction now supports four compile-time driver choices. All use a 128 x 64 logical canvas, so existing screen layouts remain compatible while users can install physically larger modules.

Driver option	Typical module	Status in this tree
OLED_DRIVER_SSD1306	0.96-inch 128x64	Available, disabled
OLED_DRIVER_SSD1309	2.42-inch 128x64	Available, disabled; uses Adafruit SSD1306 command path
OLED_DRIVER_SH1106	1.3-inch 128x64	Enabled in Config_Definitions.h
OLED_DRIVER_SH1107	Rotated SH1107 module	Available, disabled; rotation is applied at initialization

- Initialization includes a short startup delay to accommodate larger OLED modules.
- Adafruit SSD1306 and Adafruit SH110X libraries are both declared in PlatformIO dependencies.
- The selected driver must match the controller on the actual display; physical size alone is not enough to identify it.

Current build

The current configuration defines `OLED_DRIVER_SH1106`. If the installed larger display is SSD1309-based, switch the define before rebuilding.

6. AMP-mode rocker switch

A physical SPST rocker can force AMP mode during startup. The feature is controlled by `ENABLE_AMP_MODE_ROCKER_SWITCH` and currently uses GPIO 35.

Wiring and logic

Item	Configuration
Input pin	GPIO 35
Bias	External 10 kohm pull-up to 3.3 V
Switch	SPST from GPIO 35 to ground
Open	Normal boot-mode selection
Closed	LOW at boot; force <code>SPARK_MODE_AMP</code>

The switch is sampled during setup. Changing it after startup has no effect until reset. The compile-time feature also causes saved Spark-mode state to be ignored in favor of the physical boot control.

Hardware caution

GPIO 35 is input-only and has no internal pull-up. The external resistor is required. Confirm the board revision and existing wiring before attaching the switch.

7. Pedal-only unwanted-amp blacklist

The BLE scanner previously connected to the first device advertising Spark service UUID FFC0. The new logic can persist one unwanted amp's Bluetooth address and ignore it on every subsequent scan.

Block the currently connected amp

Action	Result
Hold Bank Up + Bank Down together for 2 seconds	The combination detector suppresses the individual Bank long-press actions.
Connected amp found	Its peer MAC is written to /config/BlockedAmp.config.
OLED confirmation	AMP BLOCKED plus the final portion of the address.
Automatic restart	Ignitron restarts, scans again, and ignores that address.
No amp connected	OLED shows NO AMP CONNECTED and no block is saved.

Clear the block

Hold Bank Up + Bank Down while powering on or resetting Ignitron. Keep holding until the OLED shows **AMP BLOCKLIST / CLEARED**, then release both switches. The configuration file is removed before BLE scanning begins.

Implementation details

- The raw two-button combination is detected before ButtonFever processes either switch.
- Both existing Bank long-press callbacks also check the combination, preventing tuner or FX mode from winning a timing race.
- Bank button objects are reset around the combination to avoid a delayed single/long action after release.
- The scanner prints and skips the blocked address without stopping discovery of other Spark amps.

Capacity

The current implementation stores one blocked MAC address. Clearing it restores the original behavior of connecting to any detected Spark amp.

8. Build, filesystem, and partition configuration

The project uses PlatformIO with ESP32 Arduino framework and Espressif32 platform 6.10.0. The default environment is esp32dev, while the latest verification was also performed explicitly with node32s. The firmware entry point is `src/main.cpp`, which includes the root `Ignitron.ino`.

Setting	Current value / purpose
src_dir	src - avoids a transient root .ino build artifact
data_dir	data - LittleFS content source
Filesystem	LittleFS
Partition selection	no_ota.csv
Extra scripts	download_fs.py and build_presetuuids.py
Serial monitor	115200 baud
Firmware upload	921600 baud where supported

Latest verified node32s build

Metric	Result
Build status	SUCCESS
RAM	59,588 bytes / 327,680 bytes (18.2%)
Flash	1,667,301 bytes / 2,097,152 bytes (79.5%)
Firmware output	.pio/build/node32s/firmware.bin

Deployment reminder

Firmware and LittleFS are separate upload targets. Updating firmware alone does not automatically replace preset-bank files in the filesystem.

9. Changed-file inventory and verification

File	IPT 2.0 responsibility
Ignitron.ino	LittleFS mount, serial protocol, upload state machine, rocker boot handling, blacklist clear gesture.
src/main.cpp	PlatformIO entry point including the root sketch.
src/Config_Definitions.h	Version 2.0.1, display driver selection, battery and rocker-switch configuration.
src/SparkPresetControl.cpp	Spark App preset JSON emission and preset control integration.
src/SparkDisplayControl.*	Multi-driver OLED support, mirror/tuner streams, temporary status messages.
src/SparkBTControl.*	Persistent blocked-address storage and BLE scan filtering.
src/SparkButtonHandler.*	Two-Bank-switch combination detection and conflict suppression.
src/SparkDataControl.*	Blacklist orchestration, OLED feedback, disconnect/restart behavior.
platformio.ini	src/data layout, dependencies, filesystem, board environments, helper scripts.
README.md	IPT operation and pedal blacklist instructions.

Verification performed

- Full PlatformIO dependency resolution and compile for node32s.
- Incremental rebuild after the callback race fix; build completed successfully.
- Physical test confirmed Bank Up + Bank Down blocks the connected amp after the callback hardening.
- Boot-clear behavior and alternate display controller variants should still be tested on their intended hardware.

Maintenance recommendation

This directory is not currently a Git checkout. Put it under version control before the next firmware iteration so each hardware and IPT change can be reviewed, tagged, and reproduced.

End of change record

Generated from the current IPT 2.0 firmware tree. Keep this PDF beside the matching firmware binary and filesystem image so hardware configuration and operating gestures remain traceable.